

Literature-Implied Answer to the Analytic Diagonalization Question

Run `fa_banach_001`, lane 3 — 9 August 2026

Status. `literature_implied_answer` (complete answer to Question 2.9 of arXiv:1212.5732).

Source question. Cyr–Ekstrand–Meyers–Peoples–Peters, arXiv:1212.5732v1, Question 2.9 on page 8, ask whether a Hermitian matrix family $A \in M_n(C[a, b])$ is diagonalizable over $[a, b]$ when its entries extend analytically to a complex domain containing $[a, b]$. The question refers to the three-entry notation for the 2×2 matrix in equation (1), although it writes M_n . The answer below covers both readings.

Answer. Yes, for every finite n ; moreover, the diagonalizing unitary may be chosen real analytic.

Identification. Rainer’s Theorem 1.1(A) in arXiv:1111.4475v2, page 1, gives global analytic parameterizations of the eigenvalues and eigenvectors of a one-real-parameter analytic normal family. The real-analytic class is quasi-analytic, so the finite-order-of-contact condition in that theorem is automatic. In the finite-dimensional argument, page 18 shows that normality makes the transport unitary and hence the eigenvectors can be chosen orthonormal.

For completeness, choose an open real interval I containing $[a, b]$ and contained in the complex domain of the entrywise extensions. Hermiticity on $[a, b]$ extends to I : for example, on the real axis each function

$$x \mapsto A_{kj}(x) - \overline{A_{jk}(x)}$$

is real analytic and vanishes on (a, b) , hence vanishes on the connected interval I ; the diagonal identities are identical. Thus $A|_I$ is a real-analytic Hermitian, hence normal, matrix family. Rellich–Rainer supplies real-analytic functions $\lambda_1, \dots, \lambda_n$ and an orthonormal real-analytic eigenframe $v_1, \dots, v_n$. With

$$U(t) = [v_1(t) \ \cdots \ v_n(t)]$$

we have, for every $t \in [a, b]$,

$$U(t)^*U(t) = I, \quad U(t)^*A(t)U(t) = \text{diag}(\lambda_1(t), \dots, \lambda_n(t)).$$

This is precisely continuous diagonalizability, with stronger analytic regularity.

Provenance and scope. This is an implication identified by the run, not an explicit later claim to answer Question 2.9. Rainer’s preprint was submitted in November 2011 and revised in April 2012, before the source paper’s December 2012 submission, and identifies the self-adjoint analytic case with Rellich’s classical theorem. The result fully answers Question 2.9, but does not settle the source’s separate arbitrary- n , C^1 extension problem for Theorem 2.6.

Search evidence. The run’s registry, solution, attempt, and proof-gap indexes were searched by arXiv id, exact title, “analytic Hermitian,” and “Rellich.” Exact-title and Question 2.9 web/arXiv searches found no prior run record or paper explicitly advertising itself as the answer; the decisive theorem is the direct perturbation-theoretic implication above.

References

- [1] J. Cyr, J. Ekstrand, N. Meyers, C. Peoples, and J. Peters, *Diagonalizing Hermitian Matrices of Continuous Functions*, arXiv:1212.5732v1 (2012).
- [2] A. Rainer, *Perturbation Theory for Normal Operators*, Trans. Amer. Math. Soc. 365 (2013), 5545–5577; arXiv:1111.4475v2.